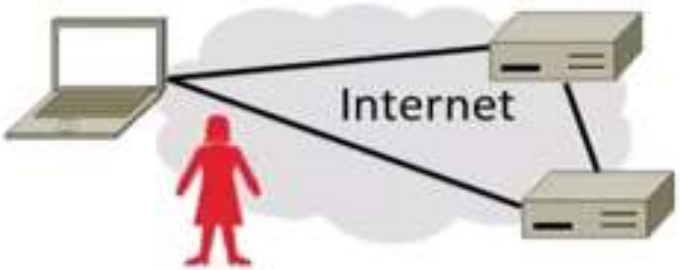
Virtual Private Network

- Logical Solution For Remote Access
- Secure Communications
- Taylored To Individuals
- Enhancing Productivity
- Increase Flexibility
- Reduce Costs Of Communications

Topic

- Virtual Private Networks (VPNs)
  - Run as closed networks on Internet
  - Use IPSEC to secure messages



Summary:

1.VPN is a method of creating a smaller sub network on top of an existing bigger network while VLAN is a subcategory of VPN

2.A VLAN is used to group computers that are not usually within same geography into the same broadcast domain while VPN is most commonly related to remote access to a company's network

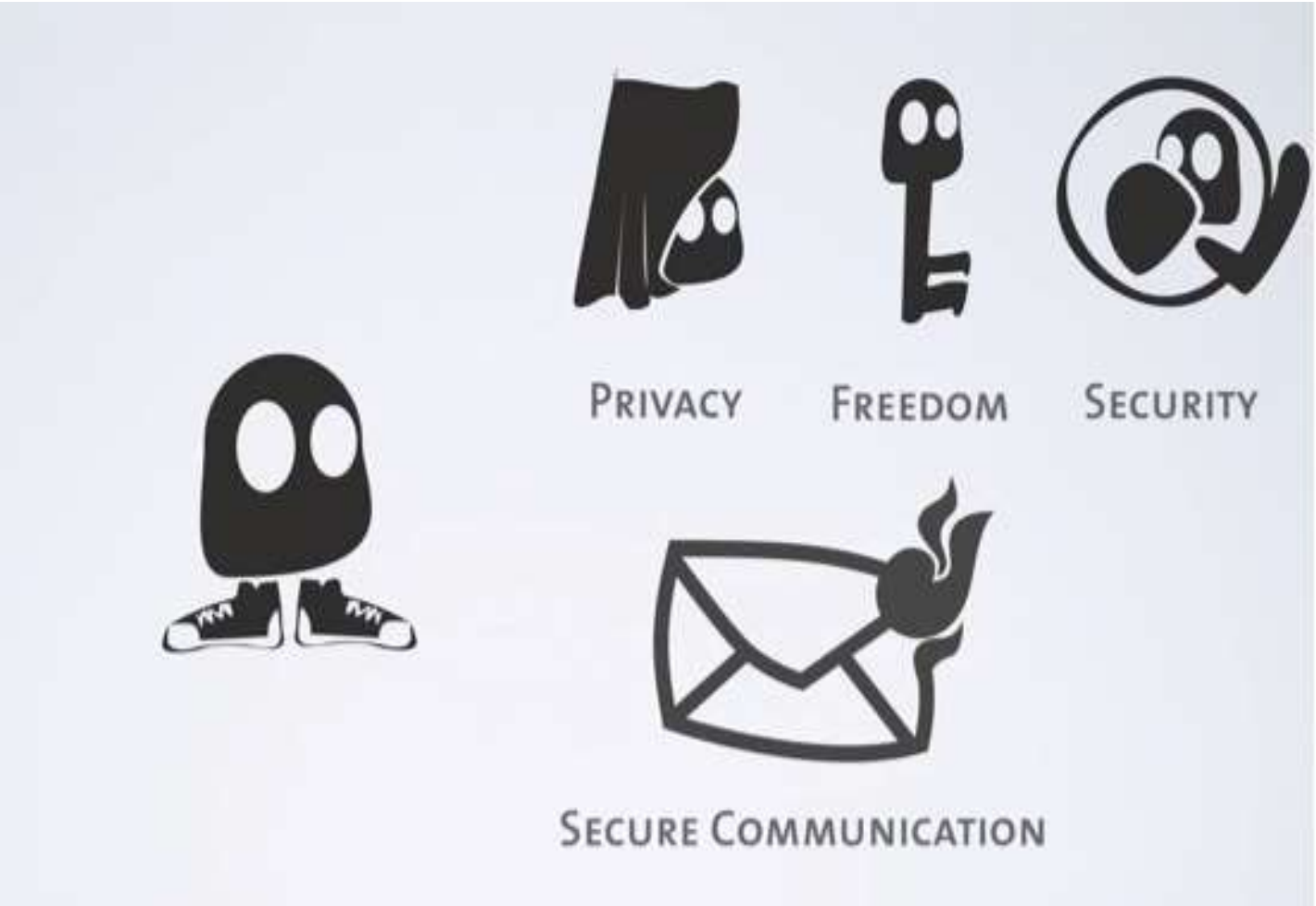
A VLAN is where you segment a local area network in to logical separate networks, each can not see the systems on the others; without specific permissions setup to do so.

A VPN (Virtual Private Network) is a wide area network (Multiple office locations for example) that is setup through a firewall to appear as one big LAN (Local Area Network). In doing so; as an example, the computers in your New York Office can see all of the computers in your Cleveland office as if they were local to New York. You can see and access the shared drives, printers, storage arrays, ect.

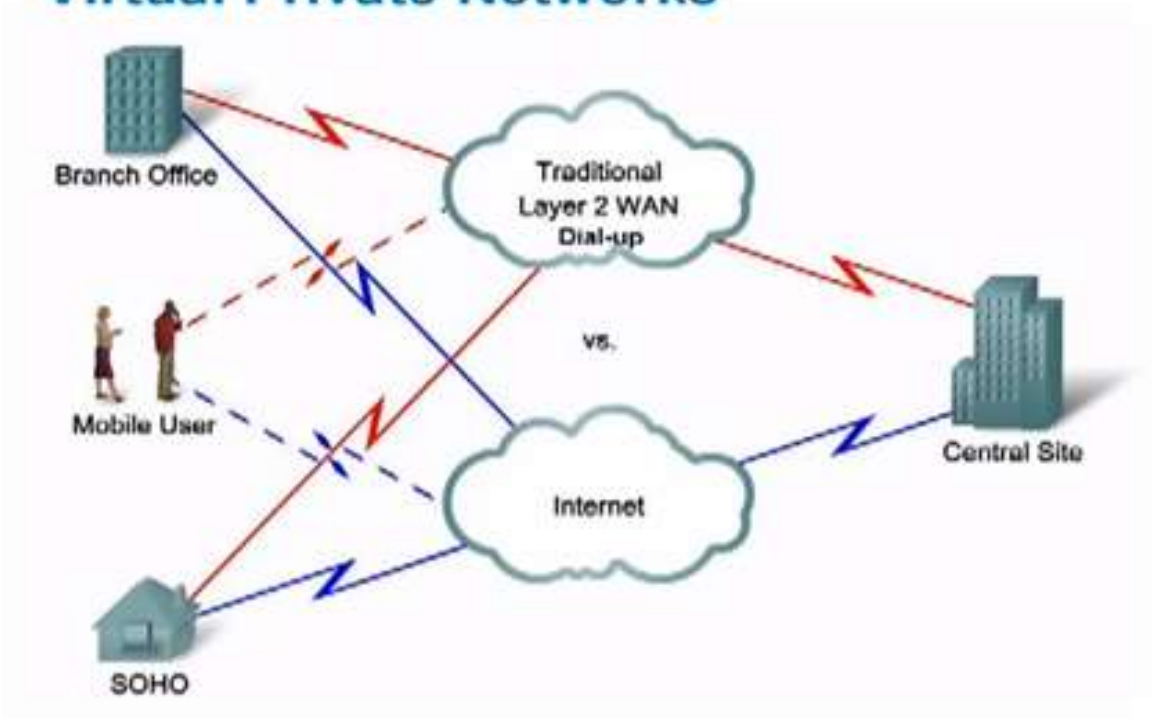
Now when I say you can see ALL devices between locations this is of course based upon the rules setup so you can LIMIT the scope of what can be accessed but you get the point.

Hope this helps

NEWS

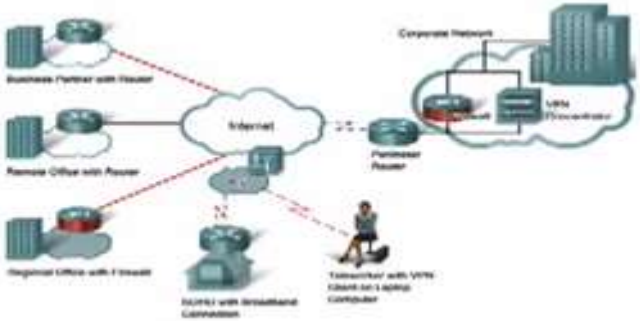

PRIVACY
FREEDOM
SECURITY
SECURE COMMUNICATION

Virtual Private Networks

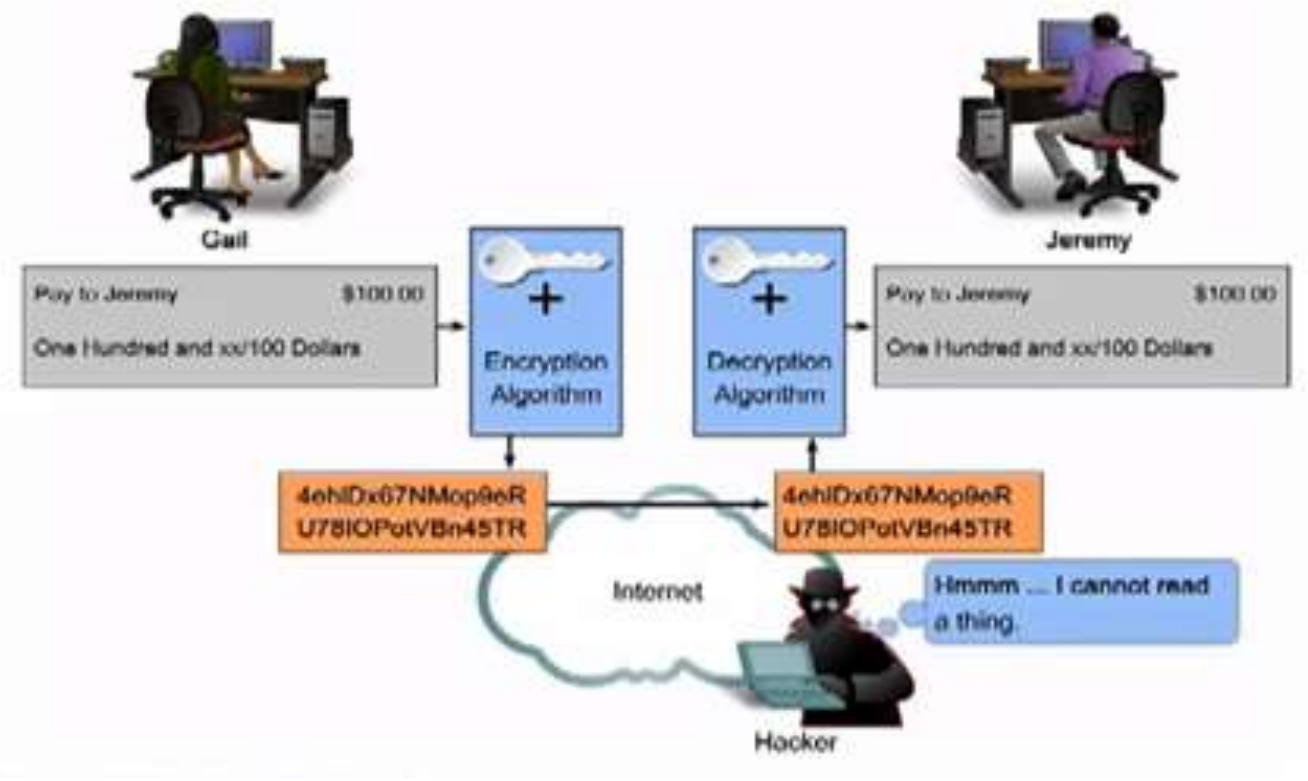


VPN Goals

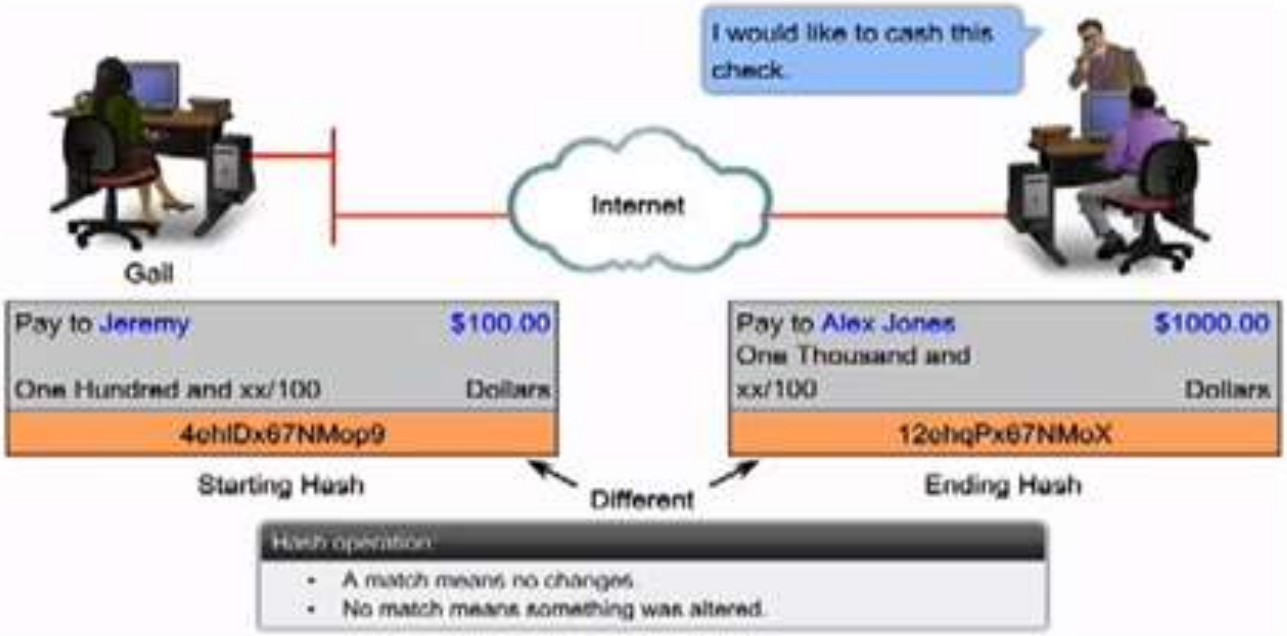
- Data Confidentiality
- Data Integrity
- Authentication



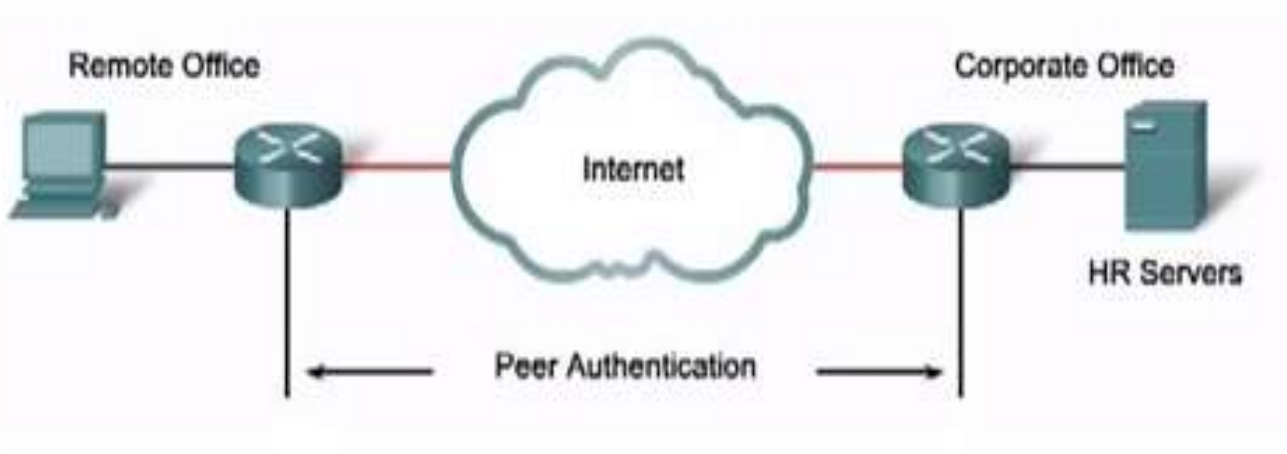
Data Confidentiality



Data Integrity

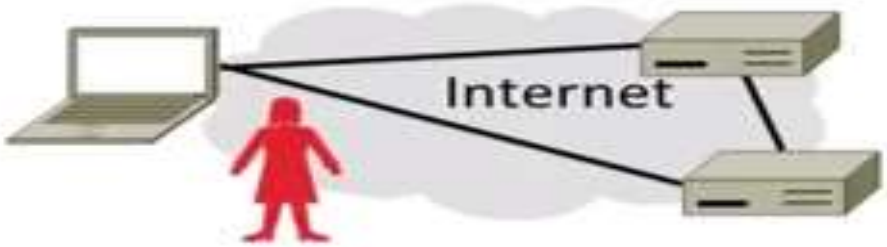


Authentication

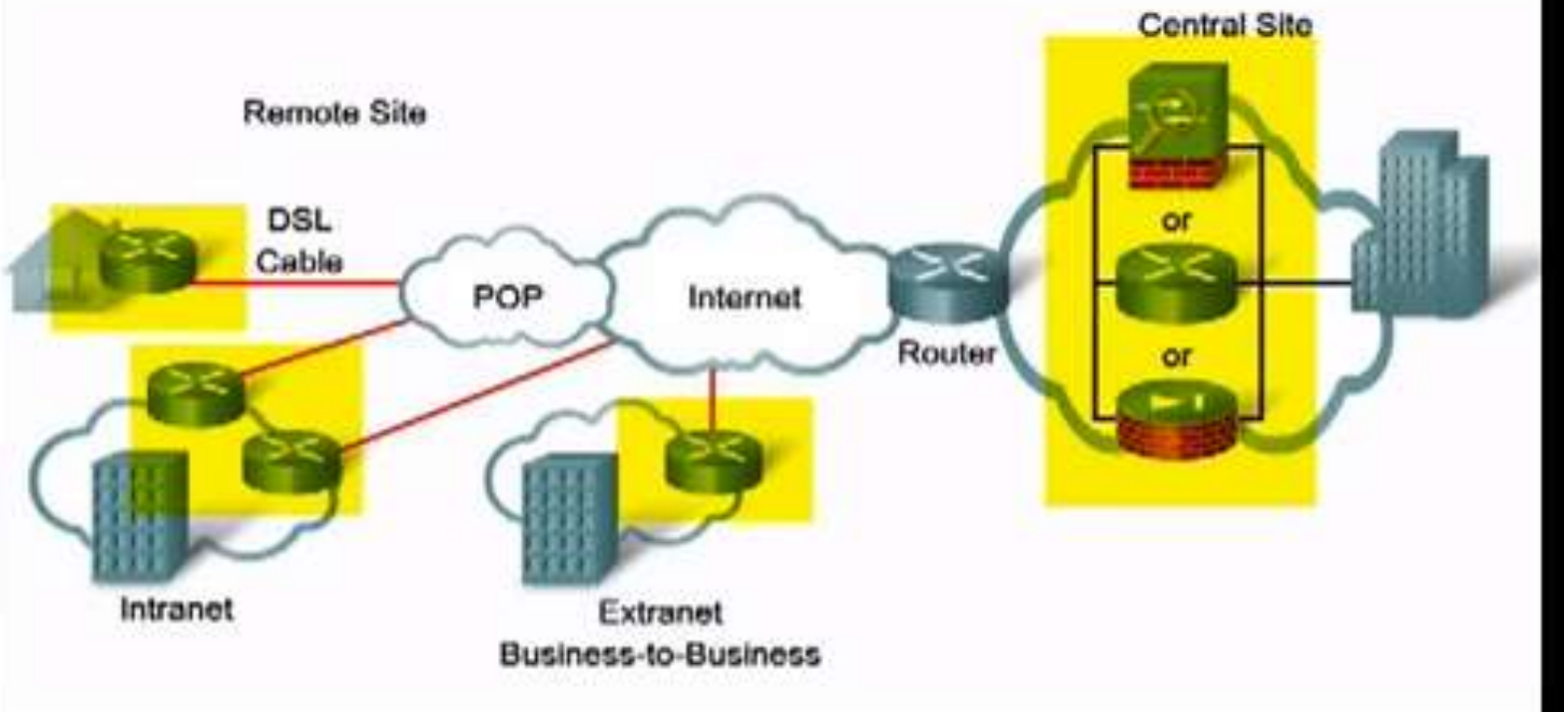


Motivation

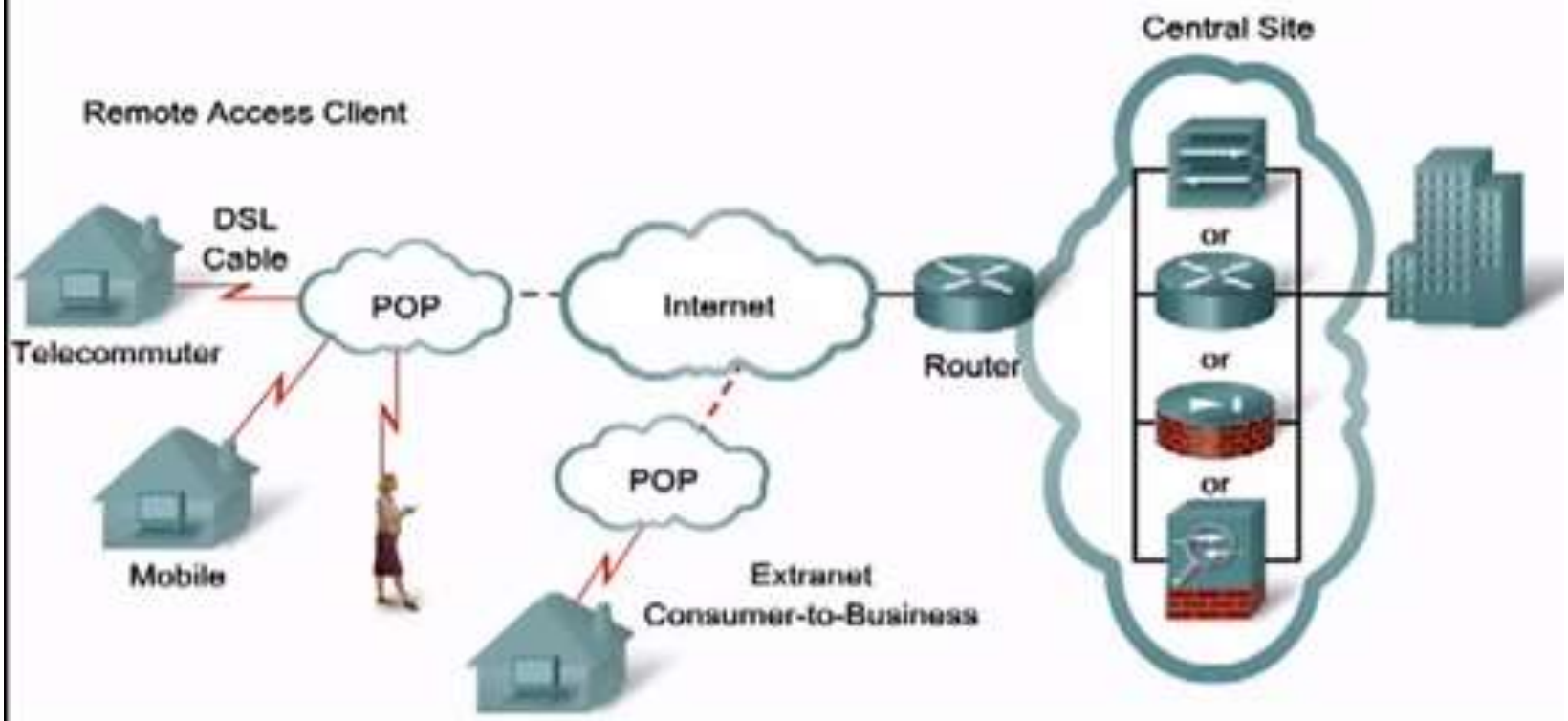
- The best part of IP connectivity
  - You can send to any other host
- The worst part of IP connectivity
  - Any host can send packets to you!
  - There's nasty stuff out there ...



Site-to-Site VPNs



Remote Access VPNs



Popular

PrivateTunnel

E-mail

Log In

Forgot your password?

Better Safe Than Sorry

- SECURITY: Secure your data to prevent man-in-the-middle attacks >>more
- PRIVACY: Protect your privacy from criminals accessing your information >>more
- PROTECTION: Protect against malware and malicious attacks >>more
- ACCESSIBILITY: Access any web content no matter where you live >>more

PROTECTION

ACCESSIBILITY

PRIVACY

Download Private Tunnel

Try it on:

Your E-mail:

Re-enter E-mail:

New Password:

Agree to the Terms of Service

Sign Up

Signup and Create an Account

Refer Your Friends

For every purchase your friend makes, you receive 10 GB of Data

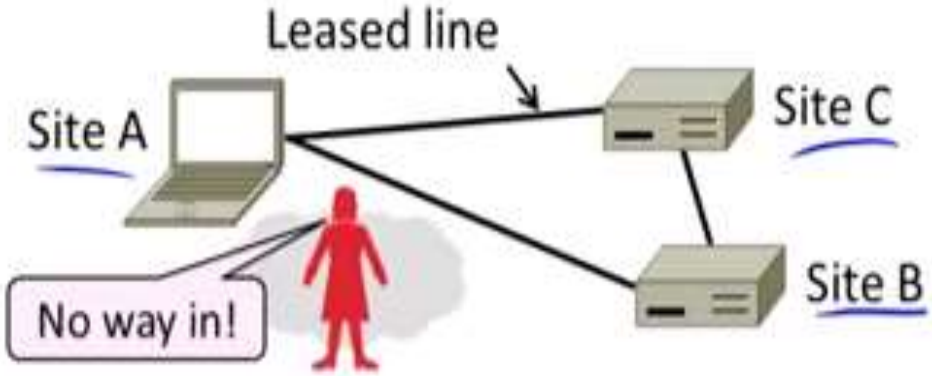
KNOW HOW...

PrivateTunnel

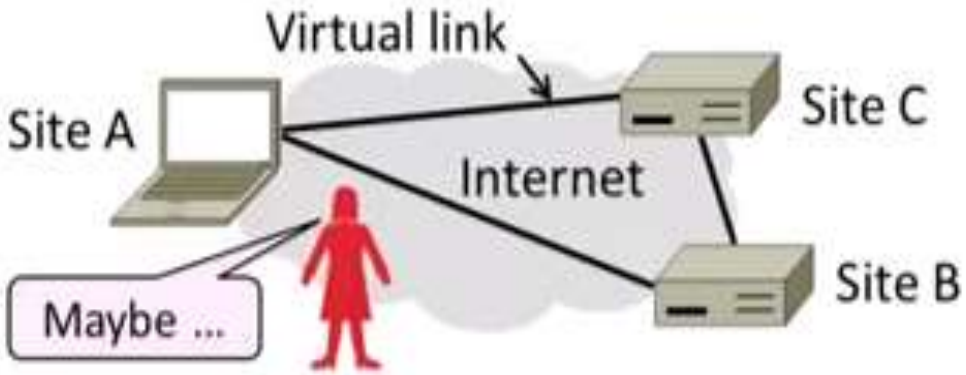
privatetunnel.com

Motivation (2)

- Often desirable to separate network from the Internet, e.g., a company
  - Private network with leased lines
  - Physically separated from Internet

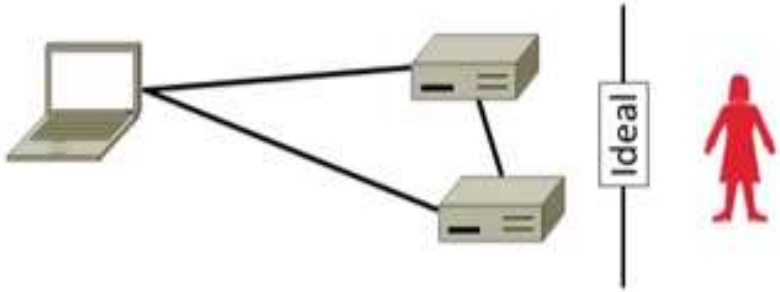


- Idea: Use the public Internet instead of leased lines – cheaper!
  - Logically separated from Internet …
  - This is a <u>Virtual Private Network</u> (VPN)



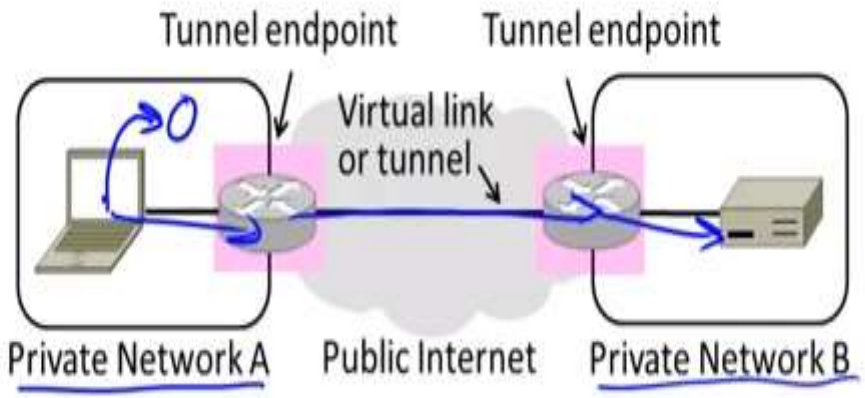
Goal and Threat Model

- Goal is to keep a logical network (VPN) separate from the Internet while using it for connectivity
  - Threat is Trudy may access VPN and intercept or tamper with messages



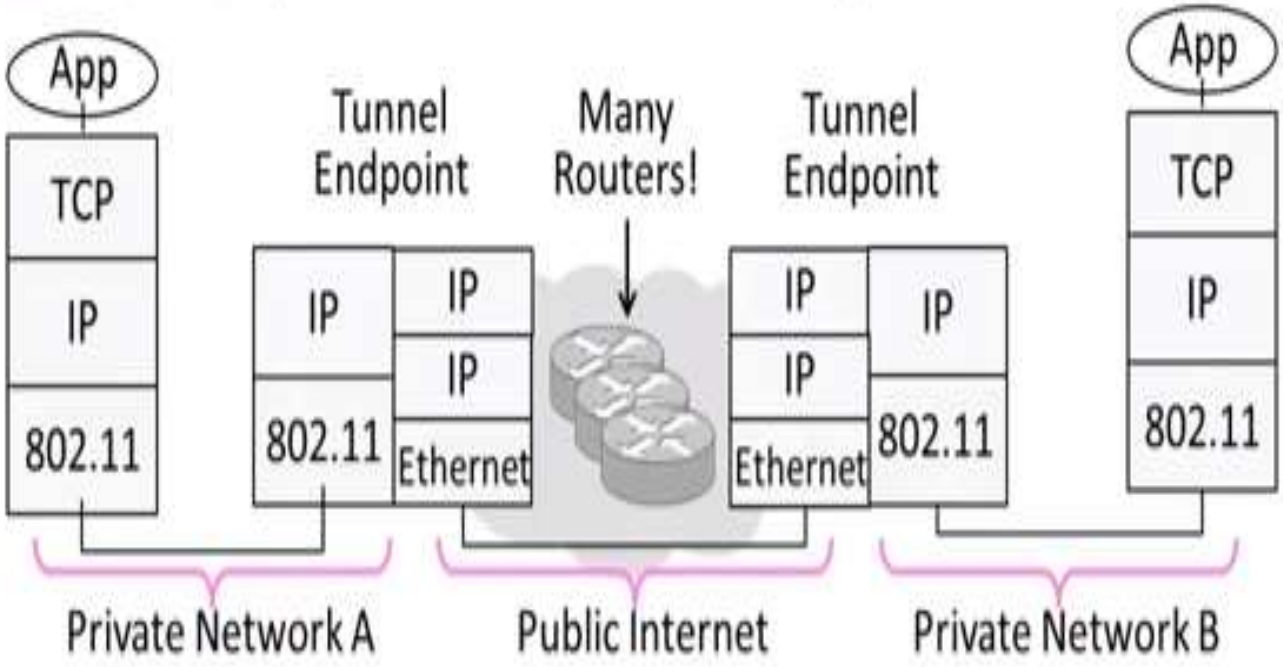
Tunneling

- How can we build a virtual link? With tunneling!
  - Hosts in private network send to each other normally
  - To cross virtual link (tunnel), endpoints encapsulate packet



Tunneling (2)

- Tunnel endpoints encapsulate IP packets ("IP in IP")
  - Add/modify outer IP header for delivery to remote endpoint



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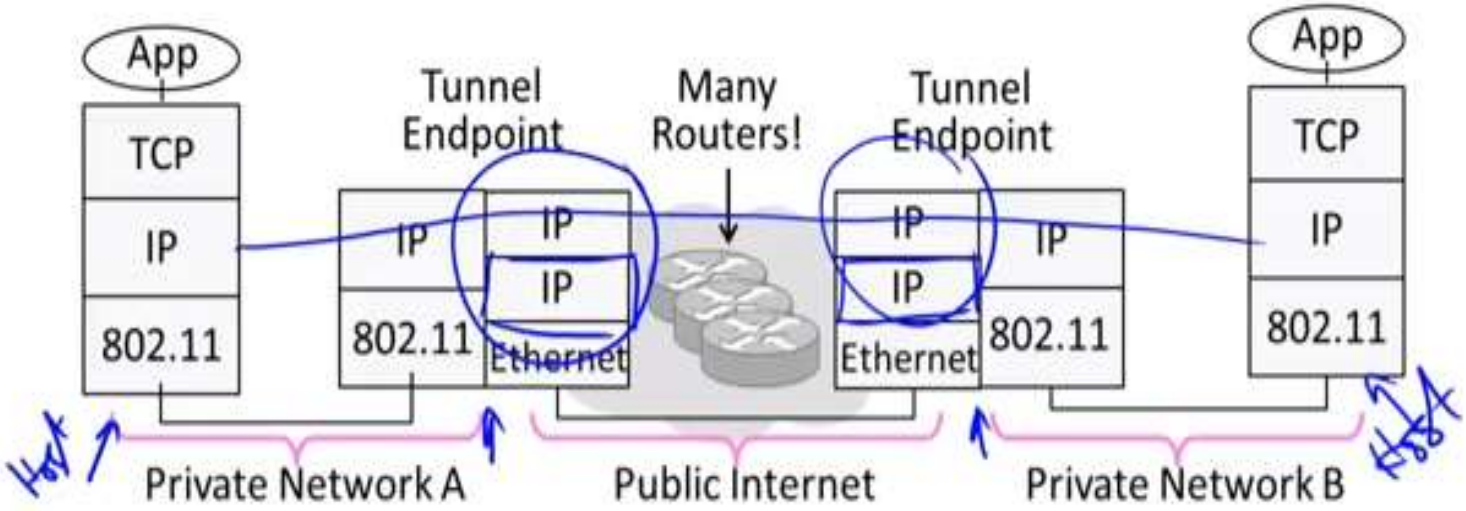
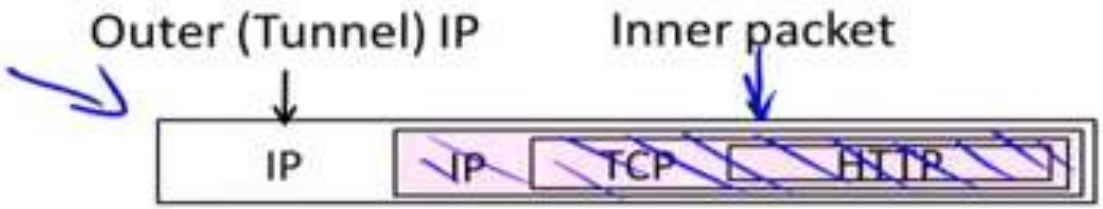


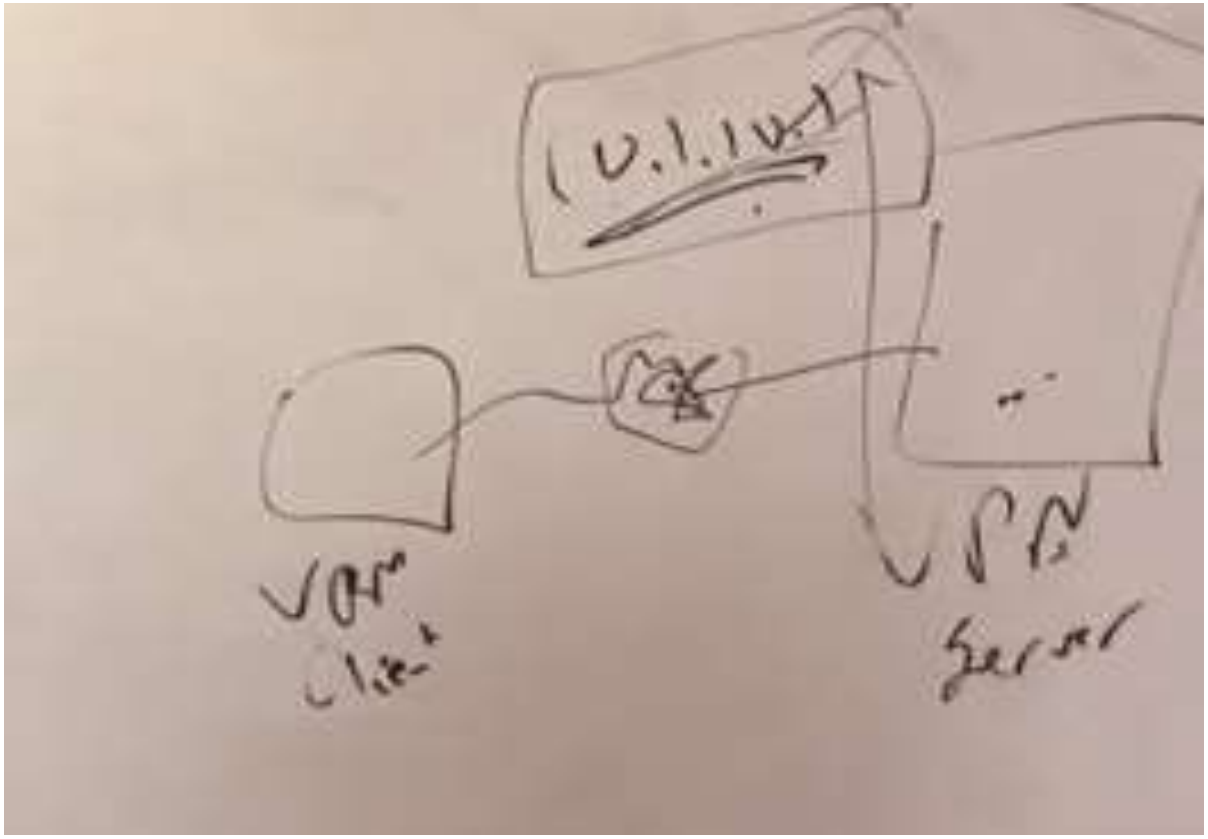
Tunneling (3)

- Simplest encapsulation wraps packet with another IP header
  - Outer (tunnel) IP header has tunnel endpoints as source/destination
  - Inner packet has private network IP addresses as source/destination



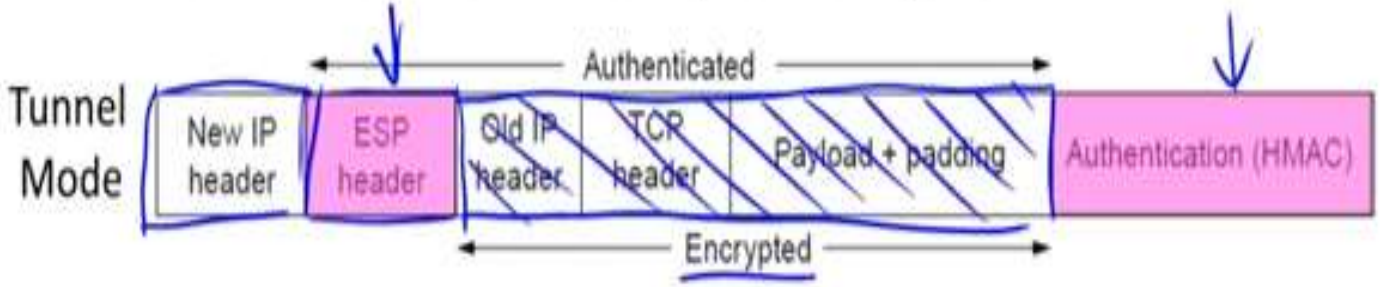
Tunneling (4)

- Tunneling alone is not secure …
  - No confidentiality, integrity/ authenticity
  - Trudy can read, inject her own messages
  - We require cryptographic protections!
- IPSEC (IP Security) is often used to secure VPN tunnels


Client
Server

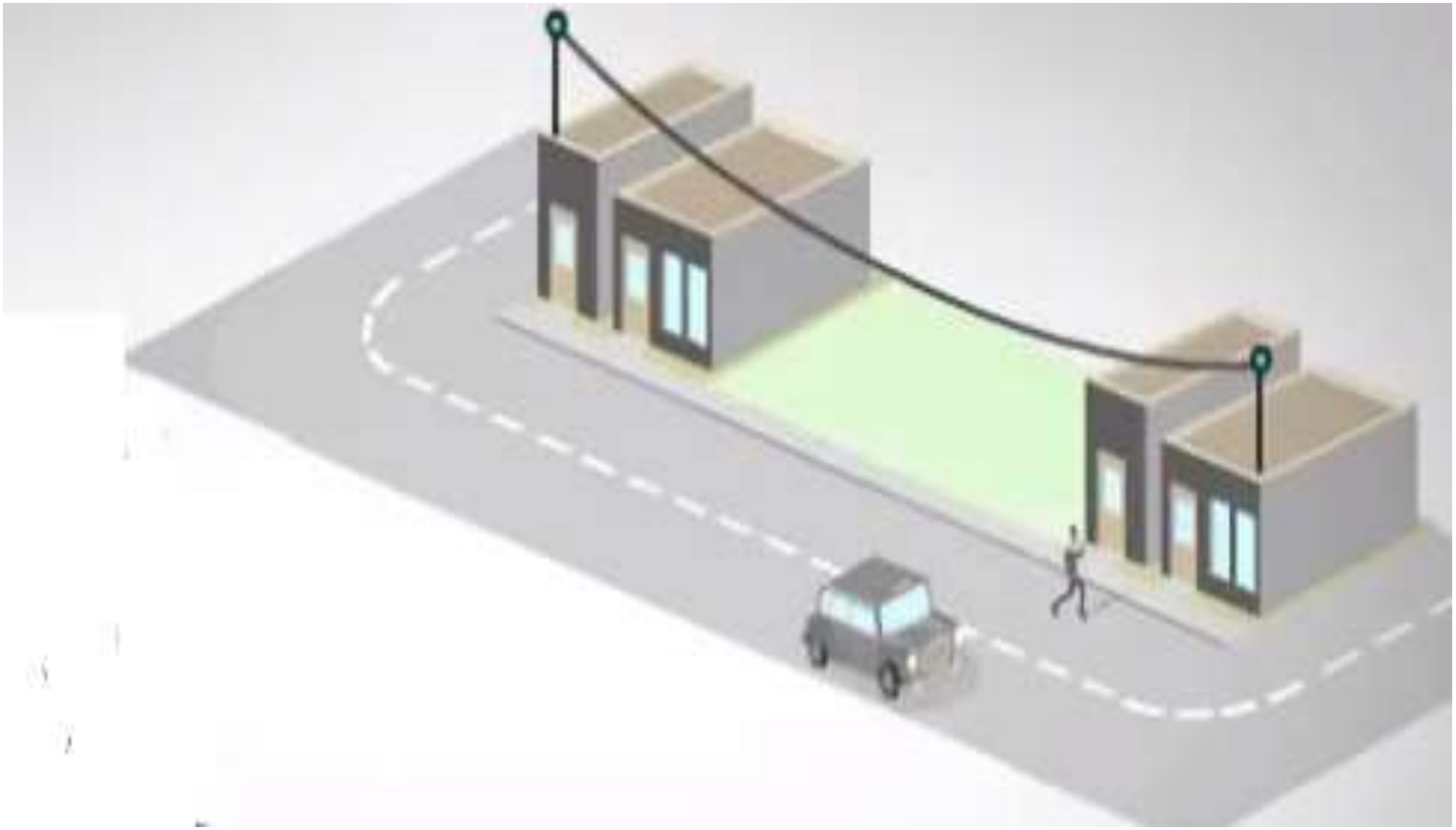
IPSEC (IP Security)

- Longstanding effort to secure the IP layer
  - Adds confidentiality, integrity/authenticity
- IPSEC operation:
  - Keys are set up for communicating host pairs → tunnel endpoint
  - Communication becomes more connection-oriented
  - Header and trailer added to protect IP packets

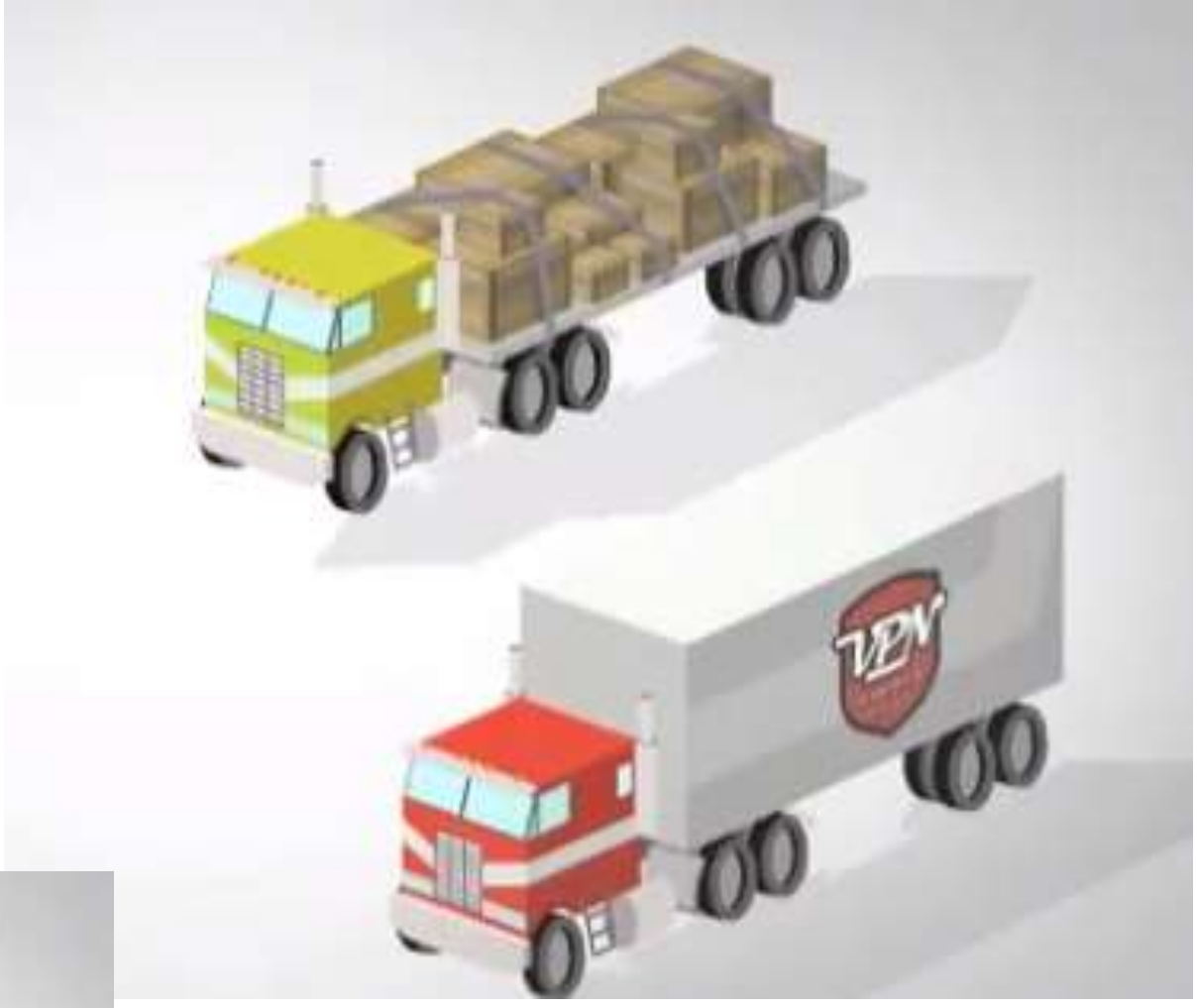


Takeaways

- VPNs are useful for building networks on top of the Internet
  - Virtual links encapsulate packets
  - Alters IP connectivity for hosts
- VPNs need crypto to secure messages
  - Typically IPSEC is used for confidentiality, integrity/authenticity

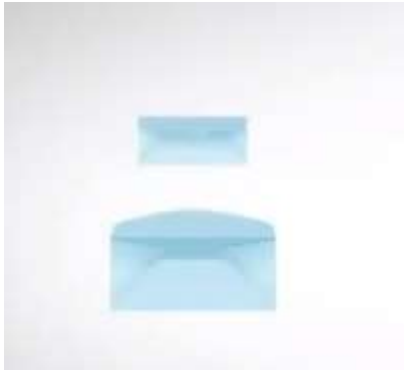


How Does VPN Differ From Other Network Connections?

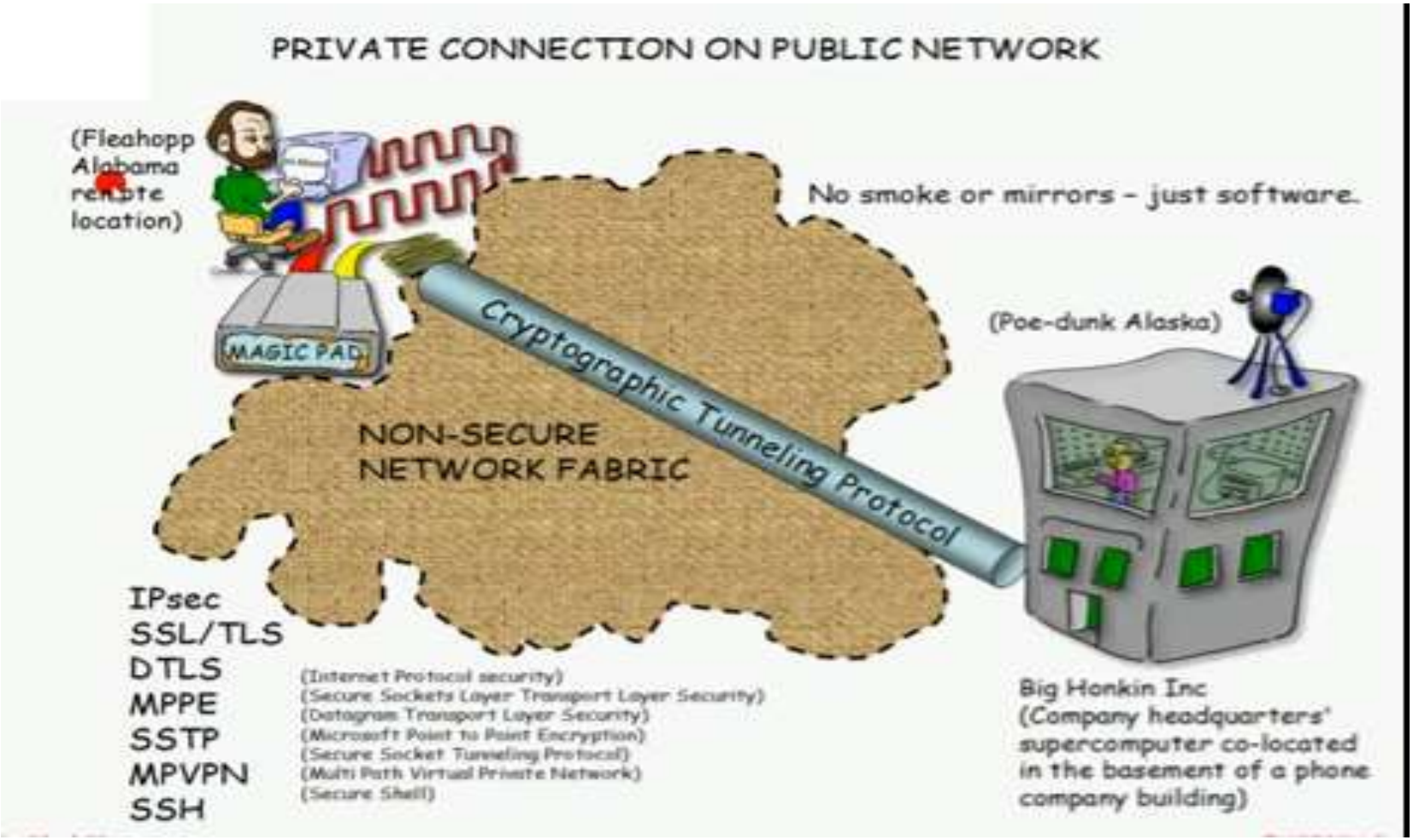


- Leased Lines Can Be Expensive
- VPN Better Utilizes Exitising Data Services
- VPN Provides Connectivity From More Locations

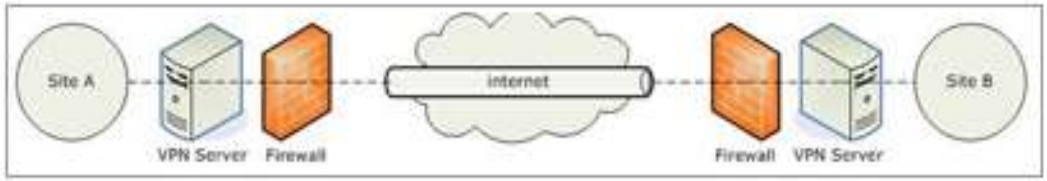
How Does VPN Tunneling Work?



- Carrier Protocol
- Encapsulating Protocol
- Passenger Protocol


PRIVATE CONNECTION ON PUBLIC NETWORK
(Fleahopp Alabama remote location)
MAGIC PAD
No smoke or mirrors – just software.
Cryptographic Tunneling Protocol
(Poe-dunk Alaska)
NON-SECURE NETWORK FABRIC
IPsec
SSL/TLS
DTLS
MPPE
SSTP
MPVPN
SSH
(Internet Protocol security)
(Secure Sockets Layer Transport Layer Security)
(Datagram Transport Layer Security)
(Microsoft Point to Point Encryption)
(Secure Socket Tunneling Protocol)
(Multi Path Virtual Private Network)
(Secure Shell)
Big Honkin Inc
(Company headquarters' supercomputer co-located in the basement of a phone company building)

BASIC CONCEPTUAL CRYPTOGRAPHIC PROCESS
CUSTOMER DATA (READABLE AT ANY NODE)
CRYPTO
X6bt)/q=V13B>#c,:+<uR1he812-^%*3M@!"v
(What goes thru the tunnel.)



VLAN vs VPN

Networks have grown astronomically over the years and has eventually led to development of the internet which spans the whole world. But having an extremely big and unsecured network means that a lot of people are able to gain access to any unsecured network and traffic. Most companies hide their local networks behind a firewall among other security mechanisms to ensure that they are safe from malicious attacks. But it is often beneficial for companies to let their employees access the company's resources from outside the local network. VPN stands for Virtual Private Network and is a method of simulating a private network that is operating on top of a larger network like the internet.

A VLAN or Virtual Local Area Network is a subcategory of VPN. Where VPN constructs range from Layer 1 to Layer 3, VLAN is purely a layer 2 construct. A VLAN makes it possible to group computers that are not necessarily connected to the same switch and make them act as if they were. VLAN makes it possible to relocate computers geographically while still maintaining the same connections and resources like shared folders and others.

VLAN can also be used to segregate computers in a bigger local network into smaller networks for each office or department and shielding the data so that they do not act as if they are on same network even if they are in the same switch.

Despite having a much larger range of applications, VPN is commonly related to technologies that allow company employees remote access to the company's resources like file and email servers. VPN also employs security mechanisms so that the data is not compromised once it goes out and travels through the internet. Users often need to authenticate their identity via a user name and password in order to gain access. Other networks also require that the device the user is on has been labeled as a trusted device before allowing access. Those who are not on a trusted device are not given access even if they have the right user name and password.

Summary:

1.VPN is a method of creating a smaller sub network on top of an existing bigger network while VLAN is a subcategory of VPN

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Now when I say you can see ALL devices between locations this is of course based upon the rules setup so you can LIMIT the scope of what can be accessed but you get the point.

Hope this helps

IF YOU HAVEN'T HAD ENOUGH YET...HERE'S SOME MORE

There's a bewildering mass of information on the web about this subject. If you have little background (or even a lot) in networking it can seem overwhelming to even begin to fully understand it. I've found several sites that appeal to my few remaining brain cells...one in particular that doesn't require B.S., M.S., PhD (Bull Sh -Moreofthe Same, Pilled higher and Deeper) to understand is here ... http://www.unixwiz.net/techtips/iguide-ipsec.html (nice piece of instruction Steve).

Here's a little excerpt:

Tunnel Mode

"Tunnel Mode forms the more familiar VPN functionality, where entire IP packets are encapsulated inside another and delivered to the destination. Like Transport mode, the packet is sealed with an Integrity Check Value to authenticate the sender and to prevent modification in transit. But unlike Transport mode, it encapsulates the *full IP header* as well as the payload, and this allows the source and destination addresses to be different from those of the encompassing packet: This allows formation of a tunnel. "

Thanks for watching! I hope this was helpful for those of you just beginning to learn about networks.